School of Engineering

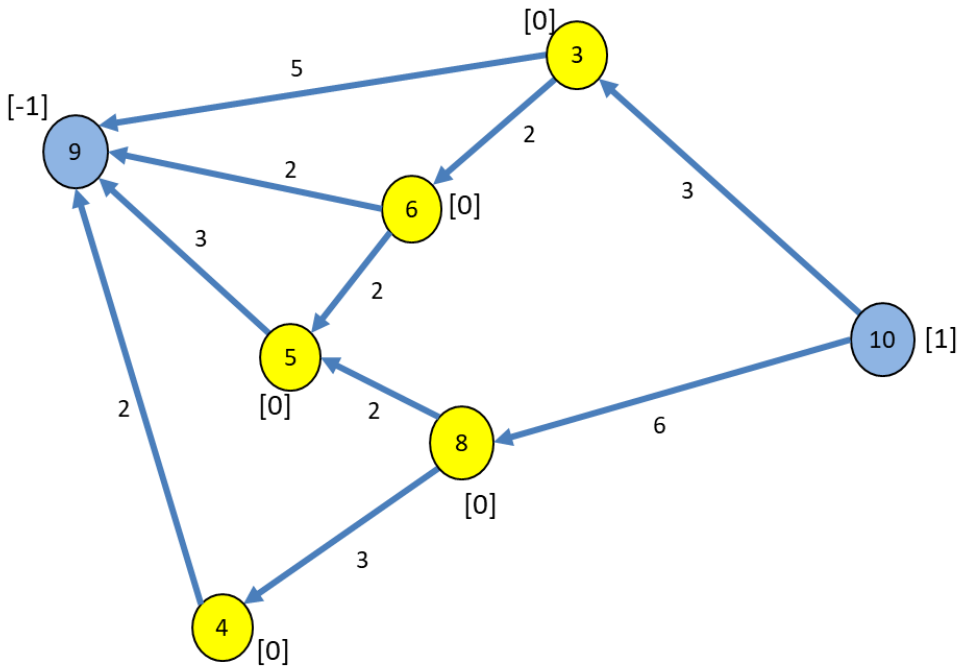
REPUBLIC POLYTECHNIC

E211 – Operations Planning II

Quiz 2 Question for Lesson 9 E-Learning

<u>Shortest Path (LP approach)</u>

Due to sudden demand increasing in Naypyidaw, WonderToys needs to quickly send a shipment of toys from Manila (10) to Naypyidaw (9) next month. The shipment has to pass through Vientiane (6) for quality inspection purposes. Figure below shows the possible shipment routes and travel time along each route (in hours).



1) Determine the respective net flow at each node.

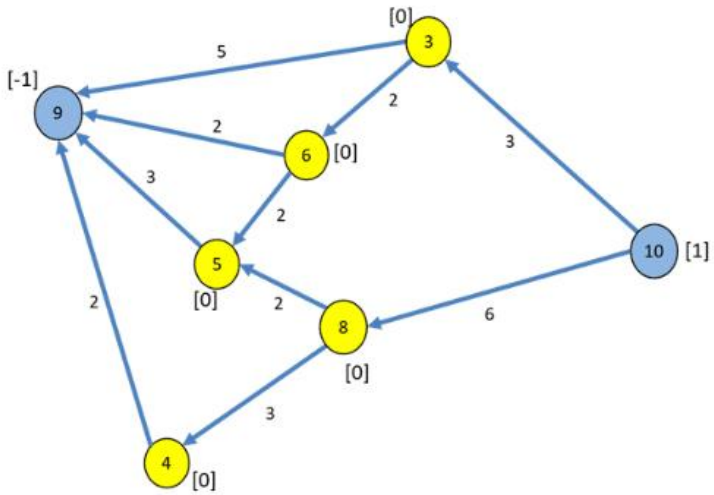
≡	Source is a supply node with	+1 unit of net flow (node 10)
≡	Destination is the demand note with	-1 unit of net flow (node 9)
≡	Transshipment is nodes with	0 unit of net flow (all other locations)

2) Formulate the decision variables. Select all answers that apply.
   - Let X_{ij} be the route from node i to node j
   - In total, there are 10 decision variables: X_{36}, X_{39}, X_{49}, X_{59}, X_{65}, X_{69}, X_{84}, X_{85}, X_{103}, X_{108}
   - X_{ij} is an binary variable

3) Write down the objective function for the shortest path LP model.
   Minimize total traveling time, $Z = 2X_{36} + 5X_{39} + 2X_{49} + 3X_{59} + 2X_{65} + 2X_{69} + 3X_{84} + 2X_{85} + 3X_{103} + 6X_{108}$

4) Construct the net flow constraints. ***Select all answers that apply.***
   Node 3: $X_{36} + X_{39} - X_{103} = 0$
   Node 4: $X_{49} - X_{84} = 0$
   Node 5: $X_{59} - X_{65} - X_{85} = 0$
   Node 6: $X_{65} + X_{69} - X_{36} = 0$
   Node 8: $X_{84} + X_{85} - X_{108} = 0$
   Node 9: $-X_{39} - X_{49} - X_{59} - X_{69} = -1$ (destination)
   Node 10: $X_{103} + X_{108} = 1$ (source)

5) Formulate Mandatory Pass constraint for Vientiane (Node 6).
   Select **two formulas**, one using outflow and the other using inflow arc(s).

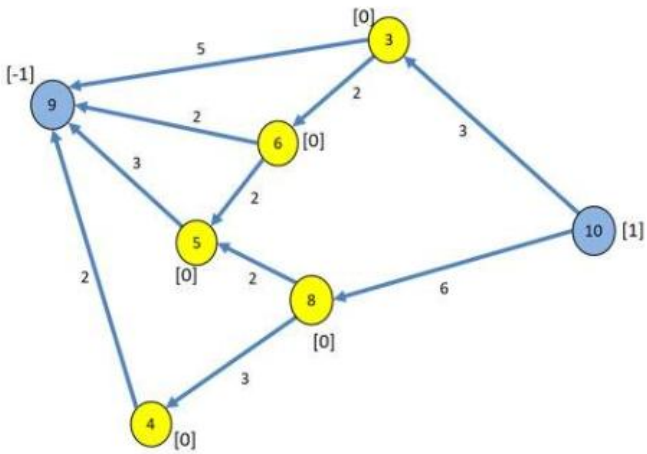


   Node 6: $X_{36} >= 1$
   Node 6: $X_{65} + X_{69} >= 1$

6) Construct non-negativity constraint
   All $X_{ij} >= 0$

7) The picture shows you the solution of the shortest path problem using Excel Solver. From the result, the route is

Decision Variables	Solution	Time (hrs)
x36	1	2
x39	0	5
x49	0	2
x59	0	3
x65	0	2
x69	1	2
x84	0	3
x85	0	2
x103	1	3
x108	0	6



Node 10 to node 3 to node 6 to node 9

8) From the result, the total traveling time is _____ hours.
(2*1)+(2*1)+(3*1) = 7 hours